Shashi Prakash

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	7.57
Foundation School , Buxar	Class XII (CBSE)	2019	75.00%
Foundation School , Buxar	Class X (CBSE)	2017	10.00%

RESEARCH WORK

Camera-LiDAR 3D Object Detection Fusion Pipeline | *Python, PyTorch, MMDetection3D, Slurm, KITTI (Mtech Thesis)*

- Debugged and re-architected a **PointPillars + CenterHead LiDAR-camera 3D detection pipeline** on KITTI (7,481 samples), running **15+ Slurm GPU jobs** with custom gradient, heatmap, and voxelization diagnostics to isolate hidden failures.
- Fixed a **nuScenes-to-KITTI decoder mismatch** that had zeroed evaluation for over a year, recovering results on **1,355/1,481 validation samples without retraining**; also corrected loss normalization (**66x reduction**) and a fusion-output wiring bug.
- Validated against an official implementation (**85.2/72.2/66.8% AP40** easy/moderate/hard), isolated the detection head to rule out model failure, and built a **geometric camera-to-BEV projection module** using per-sample calibration matrices.

SKILLS

Languages: Python, SQL

ML/DL Frameworks: PyTorch, Scikit-learn, XGBoost, LightGBM

Data Analysis: Pandas, NumPy, SciPy, statsmodels, DuckDB

ML Engineering: MMDetection3D, Optuna, SHAP, HNSW/IVF (ANN search)

Visualization & BI: Power BI, Excel

Dev Tools & Infra: Git/GitHub, Jupyter, Pytest, Ruff, Slurm (HPC/GPU scheduling)

RELEVANT COURSES

Foundation of machine learning ,Data Science Analysis, Natural language processing, Random Variable and Stochastic processes

PROJECTS

ChurnIQ --Churn Prediction & Retention Intelligence | *Python, scikit-learn, LightGBM, Optuna, FastAPI Github*

- Built a **leakage-free churn pipeline** in Python/scikit-learn for **1M customers and 32 features**; embedded preprocessing in CV and audited 4 suspect features for target leakage.
- Benchmarked **Logistic Regression vs. LightGBM** with stratified 5-fold CV, **20-trial Optuna TPE tuning**, and six dataset sizes; found similar ROC-AUC (**0.685 vs. 0.684**) but a major calibration gap **(Brier: 0.085 vs. 0.223**), using SHAP to identify key churn drivers.
- Optimized retention targeting with **isotonic calibration, decile lift, $50 offer cost, 30% save rate, and 12-month CLV**; targeting the top **10.4% by risk** captured 5,055 churners and estimated **$500K savings per 200K customers.**

India Digital Payments Intelligence | *Python, SQL, DuckDB, statsmodels, scikit-learn, Power BI Github*

- Built a **Python/SQL/DuckDB analytics pipeline** processing 150+ NPCI/RBI/Census files into an **8-table dimensional schema**; recovered **664 records** from a decade-long ETL bug and automated **10 data-quality checks.**
- Performed statistical analysis across **35 states** using Spearman, Kruskal-Wallis, and multiple regression with VIF,

heteroskedasticity, and influence diagnostics; uncovered **trend-confounded growth** and a significant negative literacy effect.
- Identified a **fintech market-sizing opportunity** by comparing transaction volume with per-capita digital intensity; found **Uttar Pradesh #3 in volume but #29 per capita**, informing state-level merchant-acquisition prioritization.

Vecsearch -- Approximate Nearest Neighbour Search from Scratch | *Python, NumPy Github*

- Implemented **HNSW and IVF ANN indexes from scratch in NumPy (1,700 LOC)**, achieving **97.7% Recall@10** with **90.9x fewer distance comparisons** on 20K 784-d Fashion-MNIST embeddings, without FAISS/hnswlib.
- Designed controlled experiments showing **metadata-correlated filters collapse ANN recall (0.664 vs. 0.998 at 50% selectivity**), exposing limitations of post-filtering and predicate-aware search on real correlated attributes.
- Optimized the ANN build hot path by replacing **~445K small NumPy calls with batched BLAS**, achieving **4.4x speedup**; validated correctness with **57 unit tests** and hardware-independent distance-count benchmarking.

Automotive IMU Sensor Fault Detection | *Python, NumPy, pandas, scikit-learn, XGBoost, SciPy, Pytest Github*

- Built an end-to-end IMU fault-detection pipeline on **2.16M samples** across 9 drives, 3 vehicles, and 3 road scenarios; vectorized feature extraction to **~60K windows/s** and shipped **34 tests with Ruff** .
- Discovered a synthetic-noise **label artifact** producing a misleading **100% F1**; fixed it using the sensor's **0.02396°C quantisation grid**, restoring **98.8% accuracy on untouched data**, and built 4 leakage-aware evaluation protocols.
- Reframed evaluation around **sensor detection limits**, showing minimum detectable faults of **12 mK flatline, 145 mK drift, and 348 mK spike**; XGBoost achieved **99.69% leave-one-vehicle-out accuracy** and feature-pruning ablation exposed model-dependent performance costs.

EXTRACURRICULAR

Represented **NIT Warangal** at Inter-NIT Basketball (2022-23); represented **IIT Hyderabad** at Inter-IIT Squash (2025) and selected for Inter-IIT Basketball (2026).

Interests: Chess, Tennis, Anime